Project Title: Telco Customer Churn Analysis

1. Introduction:

Customer churn is a major challenge in the telecom industry. This project analyzes customer behavior to identify the factors that influence churn and provides recommendations to improve customer retention.

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2. Business Problem:

The telecom company is experiencing customer loss. The objective of this project is to identify the reasons behind customer churn and provide data-driven insights to reduce churn.

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3. Tools Used:

- Excel
- Python (Pandas, NumPy, Matplotlib)
- SQL (MySQL)
- Power BI
- DAX

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4. Dataset:

- Dataset Name: **Telco Customer Churn**
- Source: Kaggle
- Number of Rows: **7043**
- Number of Columns: **21**

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5. Data Cleaning:

- Removed duplicate records
- Handled missing values
- Converted data types

- Cleaned `Total Charges`
- Exported cleaned dataset

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6. SQL Analysis

- Contract-wise churn
- Internet service analysis
- Payment method analysis
- Average monthly charges
- Average tenure

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7. Power BI Dashboard

- KPI Cards
- Customer Churn by Contract Type
- Customer Churn by Internet Service
- Customer Churn by Payment Method
- Customer Churn by Gender
- Customer Churn by Senior Citizen
- Customer Churn by Tenure Group
- Interactive Slicers

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8. Business Insights

Business Insights & Recommendations:

1. **Business Question**
2. **Observation**
3. **Business Insight**
4. **Recommendation**

1. Customer Churn by Contract Type:

Business Question:

Which contract type has the highest customer churn?

Observation:

- Customers with **Month-to-month** contracts have the highest churn.
- Customers with **One-year** and **Two-year** contracts have much lower churn.

Business Insight:

Customers without long-term commitments are more likely to leave the company.

Recommendation:

The company should encourage Month-to-month customers to move to One-year or Two-year plans by offering discounts or loyalty benefits.

2. Customer Churn by Internet Service:

Business Question:

Which internet service has the highest churn?

Observation:

- **Fiber optic** customers have the highest churn.
- **DSL** customers have lower churn.
- Customers with **No internet service** have the lowest churn.

Business Insight:

Fiber optic customers are more likely to leave than customers using other internet services.

Recommendation:

The company should review Fiber optic pricing, improve service quality, and provide better customer support.

3. Customer Churn by Payment Method:

Business Question:

Does the payment method affect customer churn?

Observation:

- Customers using **Electronic Check** have the highest churn.
- Customers using automatic payment methods generally have lower churn.

Business Insight:

Customers who pay manually may be less engaged and more likely to leave.

Recommendation:

Encourage customers to use automatic payment methods by offering incentives such as cashback or discounts.

4. Customer Churn by Gender:

Business Question:

Does gender influence customer churn?

Observation:

- Male and Female customers have similar churn patterns.

Business Insight:

Gender does not have a significant impact on customer churn.

Recommendation:

The company should focus on service quality, pricing, and customer experience rather than gender-based strategies.

5. Customer Churn by Senior Citizen:

Business Question:

Do senior citizens churn more than other customers?

Observation:

- Senior citizens have a higher churn rate compared to non-senior customers.

Business Insight:

Senior citizens may have different needs or expectations that are not being fully met.

Recommendation:

Provide senior-friendly plans, simplified billing, and dedicated customer support.

6. Customer Churn by Tenure Group:

Business Question:

How does customer tenure affect churn?

Observation:

- Customers with **0–12 months** tenure have the highest churn.
- Customers with longer tenure are less likely to churn.

Business Insight:

The first year is the most critical period for retaining customers.

Recommendation:

Strengthen onboarding, provide welcome offers, and engage new customers regularly during their first year.

7. KPI Insights:

- **Total Customers**

The dataset contains **7,043** customers.

- **Churned Customers**

Approximately **1,869** customers have churned.

- **Churn Rate**

The overall churn rate is approximately **26.54%**, meaning about one in four customers left the company.

- **Average Monthly Charges**

The average monthly charge is approximately **64.76**, indicating the typical monthly bill paid by customers.

- **Average Tenure**

The average customer tenure is approximately **32.37 months**, showing that many customers stay with the company for nearly three years.

Final Business Conclusion:

The analysis shows that customer churn is mainly influenced by contract type, internet service, payment method, and customer tenure. Customers with Month-to-month contracts, Fiber optic service, Electronic Check payments, and shorter tenure are more likely to leave. The company can reduce churn by improving customer retention programs, encouraging long-term contracts, enhancing Fiber optic services, promoting automatic payment methods, and focusing on customer engagement during the first year.

9. Recommendations:

Examples:

- Encourage customers to choose long-term contracts.
- Improve Fiber optic service quality.
- Promote automatic payment methods.
- Improve onboarding for new customers.
- Provide better support for senior citizens.

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10. Conclusion:

This project identified the major factors affecting customer churn. Using Python, SQL, and Power BI, meaningful insights were generated to support business decision-making. The recommendations can help improve customer retention and reduce churn.